MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, BIPOLAR SCHOTTKY TTL, FLIP-FLOPS,
CASCADABLE, MONOLITHIC SILICON

This amendment forms a part of MIL-M-38510/71C, dated 23 July 1984, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

TITLE: Delete and substitute as printed above.

* 1.2.3, add the following new case outlines:

"2	C-2 (20-terminal, .350" x .350"), square chip carrier package
"X	C-2A (20-terminal, .350" x .350"), square chip carrier package"

* 1.3, Thermal resistance, junction-to-case (θ_{JC}), delete and substitute the following:

"Thermal resistance, junction-to-case (θ_{JC}):
Cases A, B, C, D, E, F, 2, and X - - - - - - - (See appendix C of MIL-M-38510)"

PAGE 2

1.4: Delete "(See figure 5 for individual device type input-setup time and input-hold times)" and substitute "(See figures 5 through 16 for individual device type input-setup time and input-hold time)".

* 3.2.4, delete in its entirety and substitute the following:

" 3.2.4 Schematic circuits. Schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in the specification and shall be submitted to the qualifying activity and agent activity (DESC-ECS) as a prerequisite for qualification. All qualified manufacturers' schematics shall be maintained by the agent activity and will be available upon request."

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* The attached insertable replacement pages listed below are replacements for stipulated pages. When the new pages have been entered in the document, insert the amendment as the cover sheet to the specification.

Replacement page	Page replaced
11	11
12	Reprinted without change
19	19
20	20

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* TABLE I, delete table I (as substituted in previous amendment) and substitute the following:

" TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions 1/ $-55°C \le T_C \le +125°C$	Device	Limits Min	Limits Max	Unit
High-level output voltage	V_{OH}	V_{CC} = 4.5 V; I_{OH} = -1 mA	All	2.5 2/		V
Low-level output voltage	V_{OL}	V_{CC} = 4.5 V; I_{OL} = 20 mA	All		0.5 2/	V
Input clamp voltage	V_{IC}	V_{CC} = 4.5 V; I_{IN} = -18 mA; T_C = +25°C	All		-1.2	V
Low-level input current at D input	I_{IL1}	V_{CC} = 5.5 V; V_{IN} = 0.5 V	01	-0.5	-2	mA
Low-level input current at clear input	I_{IL2}	V_{CC} = 5.5 V; V_{IN} = 0.5 V	01	-1.5	-6.0	mA
Low-level input current at preset input	I_{IL3}	V_{CC} = 5.5 V; V_{IN} = 0.5 V	01	-1.0	-4.0	mA
Low-level input current at clock input	I_{IL4}	V_{CC} = 5.5 V; V_{IN} = 0.5 V	01	-1.0	-4.0	mA
			02,03	-2.0		
Low-level input current at J or K inputs	I_{IL1}	V_{CC} = 5.5 V; V_{IN} = 0.5 V	02,03, 04	-0.7	-1.6	mA
Low-level input current at clear input	I_{IL2}	V_{CC} = 5.5 V; V_{IN} = 0.5 V	02	-0.8	-7.0	mA
Low-level input current at clear input	I_{IL2}	V_{CC} = 5.5 V; V_{IN} = 0.5 V	04	-7.0	-14.0	mA
Low-level input current at preset input	I_{IL3}	V_{CC} = 5.5 V; V_{IN} = 0.5 V	02,03, 04	-0.8	-6.0	mA
Low-level input current at clock input	I_{IL4}	V_{CC} = 5.5 V; V_{IN} = 0.5 V	04	-4.4	-8.0	mA
Low-level input current, all inputs	I_{IL1}	V_{CC} = 5.5 V; V_{IN} = 0.5 V	05,06	-1	-2	mA
High-level input current at J, K, or D inputs	I_{IH1}	V_{CC} = 5.5 V; V_{IN} = 2.7 V	01,02, 03,04		50	µA
High-level input current at clear input	I_{IH2}		01		150	µA

See footnotes at end of table. "

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PAGE 4

* TABLE I, Propagation delay time, low-to-high level, clock to Q or $\overline{Q}$, Max Limits column: Delete "13.0" and substitute "16.0".

PAGE 7

TABLE I, footnote 2/, delete and substitute:

"2/ At T_C = +125°C, V_{IL} = 0.7 V, V_{OL} maximum = .45 V."

PAGE 8

* TABLE II, Interim electrical parameters test requirements: Delete "(pre burn-in)".

* TABLE II, Group A test requirements, class B devices: Add "10, 11".

* TABLE II, Additional electrical subgroups for group C periodic inspections: Delete in its entirety.

4.2a(1), after condition D: Add "or E".

* 4.2b., delete and substitute the following:

" b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical test parameters prior to burn-in is optional at the discretion of the manufacturer."

PAGE 15

* FIGURE 4, resistor connected to V_{CC} terminal: Delete "R" and substitute "R_1".

* FIGURE 4, resistor connected to K2 terminal: Delete "R_1" and substitute "R_2".

PAGE 17

* FIGURE 4, resistors connected to pins 4 and 14: Delete "R_{13}" and substitute "R_{11}".

PAGE 18

FIGURE 5, note 3: After t_A = 10 ns, add "±20%"; and after t_B = 10 ns, add "±20%".

PAGE 25

FIGURE 11, note 1: After 80 MHz, add "50% duty cycle"; and after 60 MHz, add "50% duty cycle".

PAGES 31 THROUGH 35

* TABLE III, pin designations, delete and substitute the following:

"

Cases															
2,X 11/	2	3	4	6	8	9	10	12	13	14	16	18	19	20
A,B,C,D	1	2	3	4	5	6	7	8	9	10	11	12	13	14
	CLR 1	D1	CLK 1	PRE 1	Q1	$\overline{Q}$1	GND	$\overline{Q}$2	Q2	PRE 2	CLK 2	D2	CLR 2	V_{CC}

"

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PAGE 31

* TABLE III, test numbers 25 and 26, Min Limits column: Delete "-1" and substitute "-0.5" (two places).

* TABLE III, test numbers 27 and 28, Min Limits column: Delete "-3" and substitute "-1.5" (two places).

* TABLE III, test numbers 29 and 30, Min Limits column: Delete "-2" and substitute "-1.0" (two places).

* TABLE III, test numbers 31 and 32, Min Limits column: Delete "-2" and substitute "-1.0" (two places).

PAGE 32

TABLE III, Subgroup 2: Delete "V_{IL} = 0.8 V" (as substituted in previous amendment) and substitute "V_{IL} = 0.7 V".

PAGE 33

TABLE III, test numbers 96 through 99, Max Limits column: Delete "10" and substitute "11.5".

PAGE 35

* Footnote 2/, delete in its entirety and substitute as follows:

"2/ Inputs: A = 2.4 V, B = 0.4 V."

* Footnote 3/, delete in its entirety and substitute as follows:

"3/ Outputs: H $\geq$ 1.5 V, L $\leq$ 1.5 V."

Footnote 6/: Delete "Q" and substitute "$\overline{Q}$".

* Following footnote 10/, add footnote 11/ as follows:

"11/ Cases 2 and X pins not designated are NC."

PAGES 36 THROUGH 39

* TABLE III, pin designations, delete and substitute as follows:

"

Cases																	
2,X 10/	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20
E,F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
	CLK 1	K1	J1	PRE 1	Q1	$\overline{Q}$1	$\overline{Q}$2	GND	Q2	PRE 2	J2	K2	CLK 2	CLR 2	CLR 1	V_{CC}

"

PAGE 36

TABLE III, I_{IL2}, tests 31 and 32, Max Limits column: Delete "-6.0" and substitute "-7.0".

* TABLE III, test number 33, Max Limits column: delete ' " ' and substitute "-6.0".

PAGE 37

TABLE III, Subgroup 2: Delete "V_{IL} = 0.8 V" (as substituted in previous amendment) and substitute "V_{IL} = 0.7 V".

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PAGE 38

TABLE III, test numbers 108 through 111, Max Limits column: Delete " " " and substitute "9.75".

TABLE III, test number 112, Max Limits column: Delete " " " and substitute "9.0".

PAGE 39

* Footnote 2/, delete in its entirety and substitute as follows:

"2/ Inputs: A = 2.4 V, B = 0.4 V."

* Footnote 3/, delete in its entirety and substitute as follows:

"3/ H $\geq$ 1.5 V, L $\leq$ 1.5 V."

* Following footnote 9/, add footnote 10/ as follows:

"10/ *Cases 2 and X pins not designated are NC.*"

PAGES 40 THROUGH 43

* TABLE III, pin designations, delete and substitute as follows:

Cases														
2,X 11/	2	3	4	6	8	9	10	12	13	14	16	18	19	20
A,B,C,D	1	2	3	4	5	6	7	8	9	10	11	12	13	14
	CLK 1	K1	J1	PRE 1	Q1	$\overline{Q}$1	GND	$\overline{Q}$2	Q2	PRE 2	J2	K2	CLK 2	V_{CC}

PAGE 41

TABLE III, test 51, Max Limits column: Delete "150 µA" and substitute "250 µA".

TABLE III, Subgroup 2: Delete "V_{IL} = 0.8 V" (as substituted in previous amendment) and substitute "V_{IL} = 0.7 V".

PAGE 42

* TABLE III, test number 110, columns J1 and J2: Delete "GND" and substitute "2.7" (two places).

PAGE 43

* Footnote 2/, delete in its entirety and substitute as follows:

"2/ Inputs: A = 2.4 V, B = 0.4 V."

* Footnote 3/, delete in its entirety and substitute as follows:

"3/ Outputs: H $\geq$ 1.5 V, L $\leq$ 1.5 V."

* Following footnote 10/, add footnote 11/ as follows:

"11/ Cases 2 and X pins not designated are NC."

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PAGE 47

* Footnote 2/, delete in its entirety and substitute as follows:

"2/ Inputs: A = 2.4 V, B = 0.4 V."

* Footnote 3/, delete in its entirety and substitute as follows:

"3/ Outputs: H ≥ 1.5 V, L ≤ 1.5 V."

PAGES 48 THROUGH 50

* TABLE III, pin designations, delete and substitute as follows:

"

Cases																
2,X 8/	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20
E,F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
	CLR	Q1	D1	D2	Q2	D3	Q3	GND	CLK	Q4	D4	Q5	D5	D6	Q6	V_{CC}

"

PAGE 48

TABLE III, I_{OS}: Add "7/". Tests 46 through 50, Max Limit column: Delete "7/" and substitute "-100" (five places).

PAGE 49

TABLE III, I_{CEX}, CLK pin, tests 52 through 57: Delete "C" and substitute "D" (six places).

PAGE 50

* Footnote 2/, delete in its entirety and substitute as follows:

"2/ Inputs: A = 2.4 V, B = 0.4 V."

* Footnote 3/, delete in its entirety and substitute as follows:

"3/ Outputs: H ≥ 1.5 V, L ≤ 1.5 V."

Footnote 7/, delete and substitute: "7/ At the manufacturers' option, I_{OS} tests 45 through 50, the following alternate procedure may be used: Apply 2.75 volts at test 45 Q1, test 46 Q2, test 47 Q3, test 48 Q4, test 49 Q5, and test 50 Q6, using min/max limits of -20/-50 mA."

PAGE 52

TABLE III, I_{OS}, V_{CC} pin tests 49 through 65: Delete "4.5 V" and substitute "5.5 V".

TABLE III, I_{CEX}, CLK pin, tests 58 through 65: Delete "C" and substitute "D".

PAGE 54

* Footnote 2/, delete in its entirety and substitute as follows:

"2/ Inputs: A = 2.4 V, B = 0.4 V."

* Footnote 3/, delete in its entirety and substitute as follows:

"3/ Outputs: H ≥ 1.5 V, L ≤ 1.5 V."

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4.4.3c(1), after condition D: Add "or E".

The margins of this amendment are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - SH, OS, TD
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5962-1063)

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Pin number			Pin name		Pin name		Pin name		Pin name		Pin name	
Case			Device type 01		Device type 02		Device type 03		Device type 04		Device type 05		Device type 06	
2 and X	E and F	A, B, C, and D	2 and X	A, B, C, and D	2 and X	E and F	2 and X	A, B, C, and D		A, B, C, and D	2 and X	E and F		E and F
1	1	1	NC	CLR1	NC	CLK1	NC	CLK1		CLR	NC	CLR		CLR
2	2	2	CLR1	D1	CLK1	K1	CLK1	K1		K1	CLR	Q1		Q1
3	3	3	D1	CLK1	K1	J1	K1	J1		J1	Q1	D1		$\overline{Q}$1
4	4	4	CLK1	PRE1	J1	PRE1	J1	PRE1		PRE1	D1	D2		D1
5	5	5	NC	Q1	PRE1	Q1	NC	Q1		Q1	D2	Q2		D2
6	6	6	PRE1	$\overline{Q}$1	NC	$\overline{Q}$1	PRE1	$\overline{Q}$1		$\overline{Q}$1	NC	D3		$\overline{Q}$2
7	7	7	NC	GND	Q1	$\overline{Q}$2	NC	GND		GND	Q2	Q3		Q2
8	8	8	Q1	$\overline{Q}$2	$\overline{Q}$1	GND	Q1	$\overline{Q}$2		$\overline{Q}$2	D3	GND		GND
9	9	9	$\overline{Q}$1	Q2	$\overline{Q}$2	Q2	$\overline{Q}$1	Q2		Q2	Q3	CLK		CLK
10	10	10	GND	PRE2	GND	PRE2	GND	PRE2		PRE2	GND	Q4		Q3
11	11	11	NC	CLK2	NC	J2	NC	J2		J2	NC	D4		$\overline{Q}$3
12	12	12	$\overline{Q}$2	D2	Q2	K2	$\overline{Q}$2	K2		K2	CLK	Q5		Q3
13	13	13	Q2	CLR2	PRE2	CLK2	Q2	CLK2		CLK	Q4	D5		D4
14	14	14	PRE2	VCC	J2	CLR2	PRE2	VCC		VCC	D4	D6		$\overline{Q}$4
15	15		NC		K2	CLR1	NC				Q5	Q6		Q4
16	16		CLK2		NC	VCC	J2				NC	VCC		VCC
17			NC		CLK2		NC				D5			
18			D2		CLR2		K2				D6			
19			CLR2		CLR1		CLK2				Q6			
20			VCC		VCC		VCC				VCC			

FIGURE 2. Terminal connections.

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Device type 01

INPUTS				OUTPUTS	
PRESET	CLEAR	CLOCK	D	Q	$\bar{Q}$
L	H	X	X	H	L
H	L	X	X	L	H
L	L	X	X	H*	H*
H	H	↑	H	H	L
H	H	↑	L	L	H
H	H	L	X	Q_0	$\bar{Q}_0$

Device type 02

INPUTS					OUTPUTS	
PRESET	CLEAR	CLOCK	J	K	Q	$\bar{Q}$
L	H	X	X	X	H	L
H	L	X	X	X	L	H
L	L	X	X	X	H*	H*
H	H	↓	L	L	Q_0	$\bar{Q}_0$
H	H	↓	H	L	H	L
H	H	↓	L	H	L	H
H	H	↓	H	H	TOGGLE	
H	H	H	X	X	Q_0	$\bar{Q}_0$

Device type 03

INPUTS				OUTPUTS	
PRESET	CLOCK	J	K	Q	$\bar{Q}$
L	X	X	X	H	L
H	↓	L	L	Q_0	$\bar{Q}_0$
H	↓	H	L	H	L
H	↓	L	H	L	H
H	↓	H	H	TOGGLE	
H	H	X	X	Q_0	$\bar{Q}_0$

Device type 04

INPUTS					OUTPUTS	
PRESET	CLEAR	CLOCK	J	K	Q	$\bar{Q}$
L	H	X	X	X	H	L
H	L	X	X	X	L	H
L	L	X	X	X	H*	H*
H	H	↓	L	L	Q_0	$\bar{Q}_0$
H	H	↓	H	L	H	L
H	H	↓	L	H	L	H
H	H	↓	H	H	TOGGLE	
H	H	H	X	X	Q_0	$\bar{Q}_0$

H = high level (steady state).
L = low level (steady state).
X = *irrelevant*.
↑ = transition from low to high level.
↓ = transition from high to low level.
Q_0 = the level of Q before the indicated input conditions were established.
TOGGLE: Each output changes to the complement of its previous level on each active transition (pulse) of the clock.
*This configuration is nonstable; that is, it will not persist when preset and clear inputs return to their inactive (high) level.

Device types 05 and 06

(EACH FLIP-FLOP)

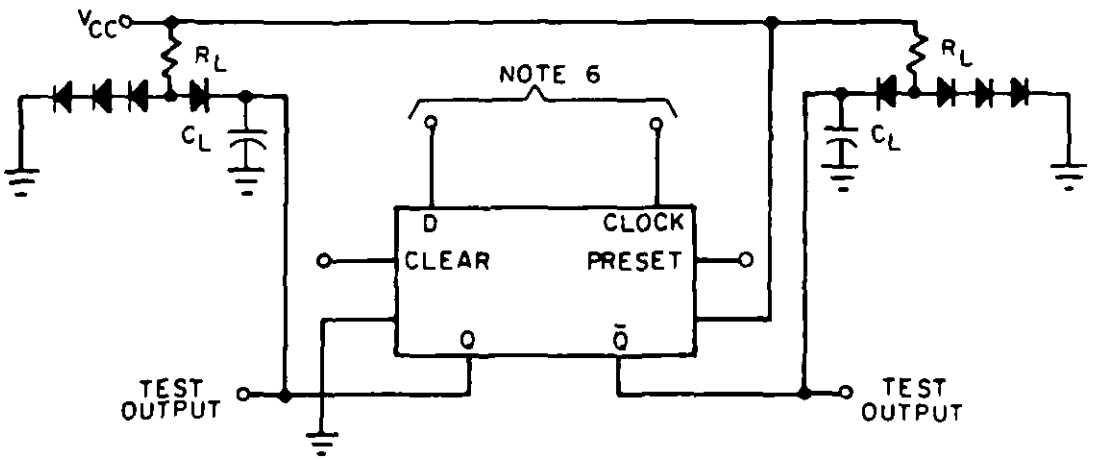
INPUTS			OUTPUTS	
CLEAR	CLOCK	D	Q	$\bar{Q}$†
L	X	X	L	H
H	↑	H	H	L
H	↓	L	L	H
H	L	X	Q_0	$\bar{Q}_0$

† = device type 06 only.

FIGURE 3. Truth tables.

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NOTES:
1. $t_r = t_f \leq 2.5$ ns, $t_{(SETUP)}$ L = 5 ns, $t_{(HOLD)}$ L = 3 ns $t_{(SETUP)}$ H = 5 ns, $t_{(HOLD)}$ h = 3 ns, PRR ≤ 1 MHz, t_D = 10 ns.
2. When testing F_{MAX} for subgroup 9, IN(D) PRR = 75 MHz with t_D = 6 ns; and for subgroups 10 and 11, IN(D) PRR = 55 MHz with t_D = 8 ns.
3. C_L = 50 pF ±10 percent, including jig and proble capacitance.
4. R_L = 280Ω ±5 percent.
5. All diodes are 1N3064, or equivalent.
6. See table III for input conditions.
7. Setup and hold time functionality may be verified by separate test from propagation delay tests, by monitoring the output at specified setup hold conditions.

FIGURE 6. Synchronous waveforms and test circuit type 01.

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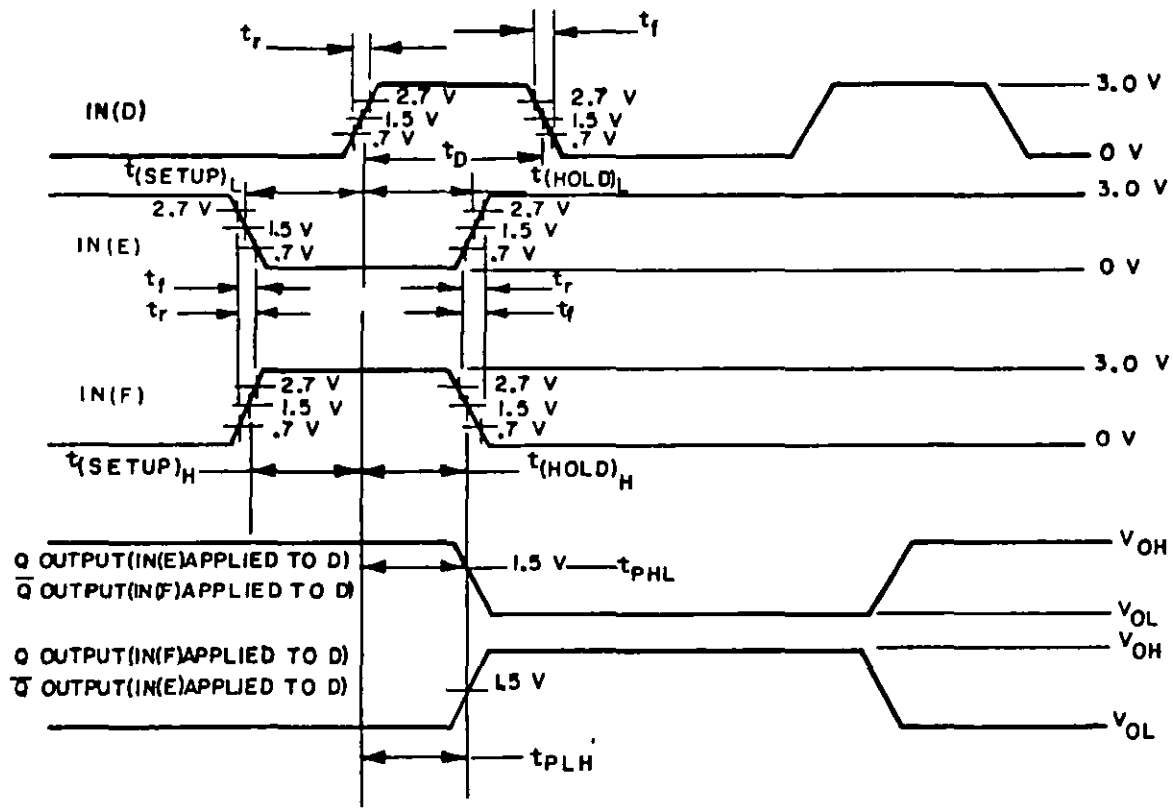


FIGURE 6. Synchronous waveforms and test circuit type 01 - Continued.

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